Permutations, Subspaces, Rank, and Dot Products

Concise revision notes from the 30 Dec handwritten matrix-operations review.

Based on handwritten MIT 18.06 revision notes.

1 Core Idea

Core idea. Early linear algebra questions often reduce to three checks: what operations preserve structure, whether a set is a subspace, and how rank controls solution behavior.

This note collects matrix-operation facts used repeatedly later: permutation matrices, subspaces, rank cases, row/column relationships, dot products, and skew-symmetry.

2 Permutation Matrices

A permutation matrix reorders rows or columns. It is orthogonal:

$$P^T P = I, \qquad P^{-1} = P^T.$$

Multiplying on the left reorders rows:

$$PA.$$

Multiplying on the right reorders columns:

$$AP.$$

Exam memory. A permutation matrix looks like the identity matrix with its rows or columns rearranged.

3 Subspace Test

A set S is a subspace when:

1. $0 \in S$;
2. if $u, v \in S$, then $u + v \in S$;
3. if $u \in S$, then $cu \in S$.

The nullspace is a subspace because if

$$Au = 0, \qquad Av = 0,$$

then

$$A(u + v) = 0, \qquad A(cu) = 0.$$

Common mistake. A set defined by $Ax = b$ with $b \neq 0$ is usually not a subspace, because it does not contain the zero vector.

4 Rank Cases

For $A \in \mathbb{R}^{m\times n}$, rank r controls solution behavior.

Rank case	Meaning	Exam consequence
$r = n$	Full column rank.	At most one solution to $Ax = b$.
$r = m$	Full row rank.	$Ax = b$ is solvable for every $b \in \mathbb{R}^m$.
$r = m = n$	Full rank square.	Exactly one solution for every b.
$r < m$ and $r < n$	Not full rank.	Zero or infinitely many solutions.

5 Row Space and Column Space

Row space and column space have the same dimension:

$$\dim C(A) = \dim C(A^T) = \text{rank}(A).$$

But they usually live in different ambient spaces:

$$C(A) \subseteq \mathbb{R}^m, \qquad C(A^T) \subseteq \mathbb{R}^n.$$

Exam memory. Same dimension does not mean the same space. Check where the vectors live.

6 Dot Product

The dot product is

$$x \cdot y = x^T y.$$

It measures alignment:

$$x^T y = 0$$

means the vectors are perpendicular.
It also interacts cleanly with matrix multiplication:

$$(Px) \cdot (Py) = x \cdot y$$

for a permutation or orthogonal matrix P.

7 Skew-Symmetric Matrices

A matrix is skew-symmetric when

$$A^T = -A.$$

The diagonal entries must be zero, because

$$a_{ii} = -a_{ii} \quad \Rightarrow \quad a_{ii} = 0.$$

Off-diagonal entries come in opposite pairs.

8 Common Mistakes

Mistake	Correct exam habit
Treating $Ax = b$ as a subspace when $b \neq 0$.	Check whether the zero vector is included.
Confusing row operations with column operations.	Left multiply changes rows; right multiply changes columns.
Assuming equal dimensions imply equal spaces.	Check the ambient spaces.
Forgetting the full-rank square case.	$r = m = n$ gives existence and uniqueness.
Putting nonzero entries on a skew-symmetric diagonal.	The diagonal must be zero.

9 Final Exam Checklist

Checklist.

1. Identify whether multiplication is on the left or right.
2. Check $P^T P = I$ for permutation matrices.
3. Test subspaces by zero vector, addition, and scalar multiplication.
4. Write the size $m \times n$ before using rank facts.
5. Find rank r.
6. Use full column rank for uniqueness.
7. Use full row rank for existence.
8. Use $x^T y = 0$ to check perpendicularity.
9. Check skew-symmetry with $A^T = -A$.